KIAN KOK MIDDLE SCHOOL
Second Semester Mock Examination (2026)

Senior Three Commerce

UEC MATHEMATICS
(Paper 1)

MARK

Date: 02 September 2026 (Wed)

(60 mins)

NAME:	NO.	CLASS:

INSTRUCTIONS TO CANDIDATES

1. This subject comprises two papers:
   Paper 1: Multiple-choice Questions (40%),
   Paper 2: Subjective Questions (60%).

2. Attempt **all** the **20** questions in Paper 1. For each question, select the **correct** answer and blacken the square corresponding to the same letter of the alphabet as the answer you have selected. Mark the answers with a **2B** or **5B** pencil on the MCQ answer sheet provided.

3. The prescribed electronic calculators may be used.

4. Mathematics Formula Sheet is on page 2-3.

5. This paper consists of 7 printed pages.

Do Not Turn Over This Page Until You Are Told To Do So

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Date : 01-09-2026	Date : **01-09-2026**	Date : **01-09-2026**

Mathematics Formula Sheet

I. Algebra

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$(a\pm b)^2=a^2\pm 2ab+b^2$$

$$(a\pm b)^3=a^3\pm 3a^2b+3ab^2\pm b^3$$

$$a^2-b^2=(a+b)(a-b)$$

$$a^3\pm b^3=(a\pm b)\left(a^2\mp ab+b^2\right)$$

$$a^m\times a^n=a^{m+n}$$

$$a^m\div a^n=a^{m-n}$$

$$\left(a^m\right)^n=a^{mn}$$

$$(ab)^n=a^nb^n$$

$$\left(\frac{a}{b}\right)^n=\frac{a^n}{b^n}$$

$$\log_a xy=\log_a x+\log_a y$$

$$\log_a\frac{x}{y}=\log_a x-\log_a y$$

$$\log_a x^m=m\log_a x$$

$$a^{\log_a x}=x$$

$$\log_a x=\frac{\log_b x}{\log_b a}$$

Inverse of a matrix $\frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$

A.P. $a_n=a+(n-1)d$

$$S_n=\frac{n}{2}\left[2a+(n-1)d\right]$$

G. P. $a_n=ar^{n-1}$

$$S_n=\frac{a\left(1-r^n\right)}{1-r}$$

II. Trigonometry

Arc length $=r\theta$

Area of a sector $=\frac{1}{2}r^2\theta$

$$\tan\theta=\frac{\sin\theta}{\cos\theta}$$

$$\operatorname{cosec}\theta=\frac{1}{\sin\theta}$$

$$\sec\theta=\frac{1}{\cos\theta}$$

$$\cot\theta=\frac{1}{\tan\theta}$$

$$\sin^2\theta+\cos^2\theta=1$$

$$1+\tan^2\theta=\sec^2\theta$$

$$1+\cot^2\theta=\operatorname{cosec}^2\theta$$

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$$

$$a^2=b^2+c^2-2bc\cos A$$

Area of a triangle $=\frac{1}{2}ab\sin C$

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III. Analytic Geometry

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Division of a line segment $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$

Area of a triangle $=\frac{1}{2}\left|(x_1y_2+x_2y_3+x_3y_1)-(x_2y_1+x_3y_2+x_1y_3)\right|$

Equation of a line $y-y_1=m(x-x_1)$

IV. Statistics and Probability

Mean $\bar{x}=\frac{\sum f_ix_i}{\sum f_i}$

Mode $=L+\left(\frac{d_1}{d_1+d_2}\right)C$

Median $M=L+\left(\frac{\frac{n}{2}-F_m}{f_m}\right)C_m$

Upper quartile $Q_3=L_3+\left(\frac{\frac{3n}{4}-F_3}{f_3}\right)C_3$

Lower quartile $Q_1=L_1+\left(\frac{\frac{n}{4}-F_1}{f_1}\right)C_1$

Interquartile range $=Q_3-Q_1$

Quartile deviation $Q.D.=\frac{Q_3-Q_1}{2}$

Variance $\sigma^2=\frac{\sum(x_i-\bar{x})^2f_i}{\sum f_i}=\frac{\sum x_i^2f_i}{\sum f_i}-\bar{x}^2$

Standard deviation $\sigma=\sqrt{\frac{\sum(x_i-\bar{x})^2f_i}{\sum f_i}}$

$$=\sqrt{\frac{\sum x_i^2f_i}{\sum f_i}-\bar{x}^2}$$

Mean deviation $=\frac{\sum|x_i-\bar{x}|f_i}{\sum f_i}$

Statistical index $I=\frac{Q_1}{Q_0}\times100$

Composite index $=\frac{\sum w_ix_i}{\sum w_i}$

V. Calculus

$$\frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$$

$$\frac{d}{dx}x^n=nx^{n-1}$$

$$\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$$

$$\frac{d}{dx}f(g(x))=f'(g(x))g'(x)$$

$$\int x^n\,dx=\frac{x^{n+1}}{n+1}+C,\quad n\neq-1$$

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1.

Find the domain of $f(x)=\frac{2x}{x^2-4}$.

A $D_f:\{x|x\in R, x\geq 2\}or[2,\infty)$

B $D_f:\{x|x\in R, x\neq \pm 2\}$

C $D_f:\{x|x\in R, x\neq 2\}$

D $D_f:\{x|x\in R, x\neq -2\}$

2.

Given that $\frac{d}{dx}\left(\frac{x}{6-2x}\right)=\frac{6}{(6-2x)^2}$, find $\int\left(\frac{3}{(6-2x)^2}\right)dx$.

A $\frac{3x}{4(3-x)}+C$

B $\frac{x}{4(3-x)}+C$

C $\frac{4x}{3-x}+C$

D $\frac{5x}{2(3-x)}+C$

3.

If $\log_{16} P=\log_4 7$,find the value of P .

A 36

B 64

C 25

D 49

4.

Given a quadratic equation $mx^2+6x+3m=0$ .Find the values of m if $-\frac{3}{m}$ is one of the roots of the quadratic equation .

A $\pm\sqrt{3}$

B $-\sqrt{3}$

C $\pm\sqrt{5}$

D $\pm\sqrt{11}$

5.

A straight line $\frac{x}{3}+\frac{y}{9}=1$, ... and $y-axis$ at N. Find the equation of the perpendicular bisector of the straight line MN.

A $x+3y+12=0$

B $3x-y+6=0$

C $x-3y+12=0$

D $3x+y-8=0$

6.

Given that $\tan\theta = \frac{1}{\sqrt{3}}$, find the value of $\frac{\cos ec^2\theta - \sec^2\theta}{\cos ec^2\theta + \sec^2\theta}$.

A $\frac{1}{4}$ B $\frac{2}{3}$ C $\frac{1}{2}$ D $\frac{3}{5}$

7.

If $A = \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix}$, find k so that $A^2 = kA - 2I$.

A 2 B 1 C 3 D 4

8.

Simplify $\frac{(a^3 - ab^2) - a + b}{(a^4 - b^4)}$.

A $\frac{a^2 + ab - 1}{(a^2 + b^2)(a+b)}$

B $\frac{a^2 - ab + 1}{(a^2 + b^2)(a+b)}$

C $\frac{a^2 - ab - 1}{(a^2 + b^2)(a+b)}$

D $\frac{a^2 + ab + 1}{(a^2 + b^2)(a+b)}$

9.

Luca has 5 pens in his pencil case. 3 of the pens are blue. 2 of the pens are red. Luca takes 2 pens at random from the pencil case. Work out the probability that Luca takes two pens of the same colour.

A $\frac{1}{5}$ B $\frac{4}{5}$ C $\frac{3}{5}$ D $\frac{2}{5}$

10.

Find the equation of the tangent to the curve $y=\frac{3}{5}x-\frac{1}{2}x^3$ at $x=-2$.

A $27x-5y+40=0$ B $27x-5y-40=0$

C $27x+5y+40=0$ D $27x+5y-40=0$

11.

Given that $\int_5^1 -f(x)dx=8$ and $\int_3^5 f(x)dx=-10$, find the value of $\int_3^1[f(x)-8]dx$.

A -15 B -18 C 20 D 19

12.

Find the quartile deviation of the data $202,252,189,248,333,99$.

A 30.5 B 31.5 C 29.5 D 41.5

13.

Find the difference between the largest and the smallest integer that satisfy the inequality $-x^2-2x+8\geq 0$.

A 6 B 7 C 5 D 3

14.

In ΔABC, $AB=9.25cm$, $AC=4.5cm$ and $BC=7.75cm$. Find the smallest angle of ΔABC.

A 29.02° B 28.02° C 27.02° D 26.02°

15.

Given that the perimeter of a sector is $12cm$ and the ratio of its area to the radius is 1:2. Find the angle of the sector .

A 15.35° B 18.15° C 12.25° D 10.42°

16.

Given the curve $y=-2(x+n)^2+m$ has a maximum point $(1,4)$, find the value of $m-n$.

A 7 B 8 C 5 D 4

17.

The function f is defined by $f:x\to\frac{1}{x^2}, x\neq 0$. Express f^n in terms of n when n is an odd integer.

A x^n B x^{2n} C x^{-2} D x^{-3n}

18.

Solve the system of equations $\begin{cases} 2x-6=y \\ 2x^2+xy-y^2=12 \end{cases}$.

A $x=\frac{8}{3}, y=\frac{2}{3}$ B $x=-\frac{8}{3}, y=-\frac{2}{3}$

C $x=\frac{5}{3}, y=\frac{8}{3}$ D $x=\frac{8}{3}, y=-\frac{2}{3}$

19.

Find the sum of all multiples of 8 between 801 and 950.

A 18675 B 15768 C 17865 D 16875

20.

Find the value of $\frac{\sqrt{7+4\sqrt{3}}}{\sqrt{9-4\sqrt{5}}}$.

A $2\sqrt{5}+4+\sqrt{15}+2\sqrt{3}$ B $3\sqrt{5}+4+\sqrt{15}-2\sqrt{3}$

C $4\sqrt{5}+4+\sqrt{15}-2\sqrt{3}$ D $5\sqrt{5}+4-\sqrt{15}+2\sqrt{3}$

END OF QUESTION PAPER